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Source: *Notes and Records of the Royal Society of London*, Vol. 63, No. 4 (20 December 2009), pp. 355-363

Published by: Royal Society

Stable URL: <http://www.jstor.org/stable/40647310>

Accessed: 30-05-2018 06:08 UTC

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JOHN FLETCHER MOULTON AND GUGLIELMO MARCONI: BRIDGING
SCIENCE, LAW AND INDUSTRY

by

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Several Fellows of the Royal Society had a role in the achievements of Guglielmo Marconi. Among them was John Fletcher Moulton. An outstanding undergraduate mathematician at Cambridge who maintained a lifelong interest in electricity, Moulton went on to become one of the most formidable lawyers practising in the London courts. His collaboration in the preparation of Marconi's first UK patent in 1897 marked the beginning of an important association with Marconi and the Marconi Wireless Telegraph Company.

Keywords: legal practice; Marconi; Royal Society; wireless telegraphy;
Nobel Prize; patent counsel

INTRODUCTION

One hundred years ago, in 1909, Guglielmo Marconi was awarded the Nobel Prize in Physics, sharing the honour with the German physicist Karl Ferdinand Braun, 'in recognition of their contributions to the development of wireless telegraphy.'¹ By then his role in the development of wireless communication had already won him public recognition, in the form of several honorary degrees and election to prestigious scientific and professional associations. More honours were to follow in later years. Election to the Fellowship of the Royal Society was, however, not among them. He was considered as a possible candidate in 1925, when his case was advanced by Oliver Lodge,² but the proposal was declined, no doubt to the disappointment of a man who was very sensitive to public recognition and would undoubtedly have warmed to the possibility of adding FRS to his name.

The reason why the candidature was unsuccessful lends itself to interesting observations on the Royal Society's criteria for the admission of new Fellows, and how these changed over time. However, this kind of analysis lies beyond the scope of the present note. My aim is rather to highlight the role of a Fellow of the Royal Society, John Fletcher Moulton, in Marconi's success story. Moulton was not the only Fellow with whom Marconi was associated in his career as an inventor and entrepreneur; in fact, his connections with persons who were either already distinguished Fellows or were to

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become Fellows in later years are well documented. Most of the Italian inventor's biographies mention Alan A. Campbell Swinton, the author of the letter of introduction as a result of which Marconi, soon after his arrival in London in 1896, obtained an interview with the powerful and always busy Chief Engineer and Electrician of the General Post Office (GPO), William H. Preece.³ In that period Campbell Swinton, a well-established consulting engineer and electrical contractor, was deeply engaged in research on electric discharges at low pressure and on X-rays; the study of these phenomena, and of their practical applications, remained one of his main lifelong interests, in recognition of which he was elected FRS in 1915.⁴

Preece too was a Fellow, elected in 1881 and known for his prodigious record of publications and endless controversies with the new generation of 'scientific electricians'. When Preece first met the young Marconi, he was himself involved in the attempt to devise a system of wireless or, as he described it, 'non-direct' communication. It was Preece who offered Marconi the technical assistance of the GPO's engineers (but not financial assistance, as he always punctiliously insisted). Moreover, his public support contributed decisively towards the rapid visibility acquired by wireless telegraphy in general and Marconi in particular.⁵

Another of Marconi's staunch supporters was Henry B. Jackson, later to become First Sea Lord. As the captain of HMS *Defiance*, a torpedo training ship, Jackson was one of the military experts who in 1896 examined the inventor's new system of communication and reported on it to the Admiralty. Three years later he was in charge of the Royal Navy's trials of Marconi's apparatus. For this task he was well qualified, having himself been involved in experiments on a very similar system. Indeed, it was for his experimental research on electricity that he was elected to the Fellowship of the Royal Society in 1901. Not only did Jackson recognize the superiority of the equipment used by Marconi, but he also strongly promoted its adoption by the Navy.⁶

Among other collaborations, much attention has been devoted to Marconi's association with John Ambrose Fleming. One of the leading experts in the industrial applications of electricity, Fleming was scientific adviser to the Edison Electric Light Company and to the National Telephone Company, as well as Professor of Electrical Engineering at University College London, and FRS since 1892. He was appointed scientific adviser to the Marconi Wireless Telegraph Co. in 1899, when preparations for the first transatlantic experiments began, and he remained with the company as an adviser until 1931.⁷

One of the scientifically gifted young men who were introduced to wireless telegraphy as employees of the Marconi Wireless Telegraph Co. was William H. Eccles. He was one of Marconi's assistants from 1899 to 1901, when he resigned to become Head of the Department of Mathematics and Physics at the South-Western Polytechnic, Chelsea.⁸

FRS AT BAR AND BENCH

Of the Fellows of the Royal Society who contributed to Marconi's achievements, one of the most important (although also one of the least studied) was John Fletcher Moulton (1844–1921). In the late nineteenth century and early twentieth century Moulton's reputation and public visibility were considerable. Lord Justice in 1906 and a life peer from 1912, he was one of the most formidable lawyers practising in the London courts at the turn of the nineteenth century.⁹ In the latter part of his life his reputation rested primarily on his role

as the first chairman of the Medical Research Committee (under the 1912 National Insurance Act) and more particularly as the head of the Committee for Explosive Services, organized by the government at the outbreak of World War I to oversee the manufacture of explosives and propellants for the British army. This work figures prominently in his obituaries and in the biography of him by his son. The energy and organizational skills with which he performed the task in very difficult circumstances (and at a time when he was heading for retirement after a busy professional life), and the success of the department, have justly earned him the attention of historians of World War I.¹⁰ Along the way, he was politically active as a member of the Liberal Party; although his parliamentary career was short, he took an active part in the work of the House of Commons and, from 1912, the House of Lords.

Biographical accounts also point to his eminence as a barrister and a judge. Between 1880 and 1914 he stood out as one of the most distinguished and successful of the London-based lawyers specializing in patent and other industrially related cases. The period was one of intense litigation, and in that fiery context Moulton's practice, as was pointed out in his obituary in *The Times*, 'grew to dimensions probably unparalleled except in the palmy days of the Parliamentary Bar.'¹¹ There was hardly any important case in which he was not involved, and usually he was on the winning side.

Moulton's educational background was outstanding. After taking a BA degree at University College London, he was admitted to St John's College, Cambridge, where he read for the Mathematical Tripos. He was soon recognized as the best mathematical student of his generation, a reputation that was confirmed in 1868, when he was bracketed Senior Wrangler with the highest marks ever gained, and first Smith's prizeman. He remained in Cambridge as a Fellow of Christ's College until 1873, when he moved to London to read for the Bar. He was admitted to the Middle Temple in 1874, and from there moved on to what was, from the start, a brilliant and highly remunerative career, initially as a common law junior and from 1885 as a Queen's Counsel.

The passing of the new Patent Act in 1883 brought about a marked increase in the number of patents, and the instances of actions for infringement increased accordingly. In the last quarter of the nineteenth century, often lengthy legal battles raged in all sectors of industry, but they were particularly violent in the new, rapidly developing sectors of the chemical and electrical industries. Moulton thrived in court duels with legal opponents and in cross-examinations of expert witnesses. These trials gave him ample opportunity for displaying his professional skills: contemporaries highlighted his capacity to absorb large amounts of detailed information, his oratorical powers, his ability in identifying successful lines of attack, allied to his effectiveness in the difficult art of cross-examination at a time when the experts who were called in were often some of the most distinguished scientists and engineers.

From 1885 to 1906 he was involved in the prolonged (and hugely expensive) bouts of litigation for the control of the telephone and incandescent lamp industries, and in the rapidly expanding field of the production and distribution of electricity. He was leading counsel for the United Telephone Co. and later of the National Telephone Company in their numerous cases against infringers; and he appeared regularly as a member of the legal team of the Edison & Swan United Electric Light Co. He was also involved in several cases relating to the chemical industry, most notably in connection with the Dunlop and the Welsbach patents. His defeats were as memorable as his victories; the most famous was on the occasion of his acting as counsel to Hiram Maxim in the cordite

case. This was a case brought against the General of Ordnance factories at Woolwich, representing the British Government. In 1897 the judgment went against Maxim, so allowing the government to eliminate a dangerous obstacle to the use of nitroglycerine in the manufacture of explosives. Moulton mounted a formidable case, but on this occasion it was insufficient to counter the exceptional array of legal and scientific experts called by the government on behalf of the defendants.¹²

Moulton's performances in court have been described, albeit in a rather anecdotal way, in several contemporary publications—journals, newspapers and biographies of contemporaries. The descriptions of him were often hyperbolic: a giant, a superior mind, brilliant, formidable. One of the most vivid, and enthusiastic, accounts was offered by the wife of John Hopkinson, a famous engineering consultant who was extensively engaged as an expert witness in cases relating to the electrical industry. Evelyn Hopkinson assiduously attended some of the most famous court hearings and described Moulton vividly:

It was arresting to see him, tall and distinguished looking, stand up in his seat, putting it back with something of a bang, and beginning a great cross-examination or speech. There was something of the Roman patrician about him and the way he stood up and adjusted his long black gown. . . . A perceptible thrill would pass through the court; he was the man of the moment.¹³

However, Moulton's public image and reputation are best conveyed by the cartoon by 'Spy' (reproduced as the frontispiece to this issue of *Notes and Records*), which shows him as an inquisitive, forward-leaning figure, looking straight into the eyes of an invisible counterpart.

Moulton's intense professional activity did not prevent him (at least until the early 1880s) from engaging in scientific research, especially in the field of electricity. And it was his work on electric discharges in rarefied gases, performed with William Spottiswoode, that earned him admission to the Fellowship of the Royal Society in 1880. The collaboration between Moulton and Spottiswoode resulted in a series of articles on the subject, published in *Proceedings of the Royal Society* and other journals between 1879 and 1883.¹⁴ But after Spottiswoode's death in 1883, with his professional commitments becoming more and more strenuous, Moulton's involvement in research ceased. Nevertheless, to the end of his life he maintained a keen interest in scientific and technological matters, especially in aspects that bore on industrial practice. Electrical technology was one of the sectors that he followed most closely, especially in the last two decades of the century¹⁵ (figure 1), although in later years his interests broadened to include other fields, notably medicine and chemistry. Moulton was also deeply interested in education and was an active member of the Senate of the University of London.¹⁶

Moulton's educational background, his scientific affiliations and his professional activity gave him the opportunity of establishing an extensive network of connections with some of the most eminent scientists, engineers and industrialists of his time—among them Kelvin, Rayleigh, Charles Siemens, Hiram Maxim, John Hopkinson, Frederick Bramwell, William Ayton, James Swinburne and William Preece. Such men had often been questioned in the witness box by their friend and fellow-FRS—whether they were on his or the opposite side. It is a tell-tale indicator of his reputation that on several occasions, when these eminent persons (or their biographers) wanted to boost their skills and competence as expert witnesses, they referred to their capacity to withstand Moulton's



Figure 1. In 1893 Moulton attended the World's Columbian Exhibition and associated electrical congress in Chicago. He is seen (standing, back row right) with a party of electrical engineers visiting the Niagara Falls hydroelectric power station. The group consists of F. Jaynes, W. H. Preece FRS, Moulton (back row, standing, left to right); Col. R. C. Clowry, Professor G. Forbes FRS (displaying the plan of a 5000-horsepower alternator for the Niagara Falls scheme), Major H. C. L. Holden RA FRS, Major A. H. Bagnold RE (centre, seated); and Capt. A. J. Saltren Willett RA (on the rug). Jaynes and Clowry were both associated with the Western Union Telegraph Company. Bagnold and Holden were present at the exhibition on behalf of the War Office, Preece on behalf of the Post Office Telegraph Department. (Image courtesy of IET Archives.)

cross-examination or to provide him with inputs beyond the reach of his proverbial powers of memorizing and assimilating information.¹⁷

MOULTON AND MARCONI

Along with his legal practice, Moulton was often consulted in negotiations for the purchase or licensing of patents, and he assisted patentees (at least those who could afford his fees) in the preparation of specifications, advising them as to the validity of their patents. It was in this capacity that he first encountered Marconi.

In 1897, at which time the final specification of his first patent had not yet been submitted, Marconi was approached by people interested in the purchase of his invention. Moulton was engaged by the prospective buyers to assess the commercial value of the yet-to-be-obtained patent on the basis of the description provided by the inventor. Marconi was well aware not only of the potential value of his new system of wireless communication but also of the difficulty of obtaining a sound patent for a very complex invention—sound enough to

withstand examination by those who had already declared their intention of challenging its validity. In the attempt to maximize his chances of obtaining such a patent, Marconi was already availing himself of the assistance of qualified patent agents and lawyers; he was also counselled by an experienced patent expert, John Cameron Graham. In the event, the prospective buyers abandoned the negotiation. But Graham suggested to his client that he should retain the services of the famous barrister who had been called as an examiner. So it was that Moulton became Marconi's counsel in the preparation of the final specification of his first patent.¹⁸

Moulton's presence among the experts who assisted Marconi in preparing the patent of 1897 did not escape the attention of contemporary observers, especially those who were ready to question its validity as soon as its content became public. Writing to his friend and colleague Oliver Lodge, Silvanus Thompson observed:

I happen to know that Moulton was called in to advise Marconi on the claims of his final specification of patent . . . and he advised him to claim *everything*. I understand that as the claims are drawn, they claim, *for telegraphy*, not only coherers, oscillators, and such like details, but even Hertz-waves!¹⁹

Judging from the results, Moulton's contribution was very effective. As *The Electrician* observed with more disappointment than admiration, the final specification was 'a model of perspicuity'.²⁰ Other comments were less diplomatically formulated; Oliver Lodge described it as an example of 'extreme, not to say diabolical ingenuity'.²¹ In both cases it is not difficult to imagine that the authors of comments of this kind were referring not so much to Marconi as to his advisers, and that the 'diabolical' quality of the text was being ascribed especially to Moulton.

The fact that Moulton was one of the advisers was also a guarantee. It is not surprising, then, that when the Marconi Wireless Telegraph Co. was created in 1897, it was Moulton who wrote one of the reports on the patent that were printed in the prospectus, as an assessment of the company's main asset. In the document he reassured prospective shareholders that 'when granted upon the complete specification and claims as settled by me, [the patent] will be good and valid.'²² Coming from such an authoritative lawyer, no declaration could have given more effective support to the new company.

Without detracting from Marconi's merit, there is no doubt that the foundational value of his first patent owed a great deal to the skills and expertise of his advisers, and especially to Moulton, whose role was decisive. The contribution of this powerful team was privately acknowledged by Marconi.²³ Such expertise did not come cheaply, and although no detailed records of the payments to Graham and Moulton have survived, the fees must have been high. As is pointed out in his *Oxford dictionary of national biography* entry, 'there were not many barristers of his time who earned more than he did.' Costly as Moulton certainly was, Marconi and the Marconi Wireless Telegraph Co. retained him as counsel, to work (along with a highly qualified team of patent experts) not only on the preparation of subsequent patents but also on the assessment of rival patents and the fashioning of the company's patent policy.

In collaboration with Graham and the patent agents Carpmael & Co. (also with J. A. Fleming, after he was appointed scientific adviser to the company in 1899), Moulton was consulted on applications for German patents, most notably in dealing with the rival patents of Ferdinand Braun.²⁴ He was also consulted on the legal problems faced by Marconi's patents in the USA. In 1904, when Marconi brought a suit for infringement

against the De Forest Wireless Telegraph Co., Moulton examined and reported on the publications that were claimed to have preceded Marconi's invention and were alleged to describe methods for the communication of intelligence across space.²⁵ Most importantly, he contributed to the preparation of the patent for the tuning device known as '7777', filed in 1901. Confirmation of the validity and strength of this patent were crucial if the Marconi Wireless Telegraph Co. were to counter the challenge of Lodge's patent for a syntonic device.²⁶ In the end, the only way of meeting the challenge was to settle out of court, by negotiating the purchase of the patent.²⁷ However, there is no doubt that a great deal of expertise and competence in the formulation of the specification allowed those patents, the '7777' in particular, to respond at least to some extent to the competition from rival technologies.

BEYOND THE MARCONI CASE

So what more might we hope to know about Moulton? I simply do not know to what extent Moulton has been studied by legal historians. Then it is certainly the case that his scientific achievements do not offer much scope for historical analysis, if narrowly interpreted. And there are significant obstacles when it comes to dealing not only with Moulton but also with other barristers and judges who, like him, were involved in important decisions about the organization and development of British industry. The main difficulty here is the paucity of documentary sources. Moulton did not preserve his papers, either professional or private,²⁸ so the study of his contributions has to depend on information drawn from secondary sources, such as the extensive legal reports in contemporary newspapers and technical journals, or from references to his work in the records of the individuals and companies for which he acted as consultant and adviser. However, even in relation to Marconi the information to be gleaned is scanty; in the recently recatalogued Marconi Archive, now in the Bodleian Library, Oxford, there is no trace of direct correspondence with Moulton. What limited information we have, therefore, opens no more than a narrow window, albeit one that suggests a promising, if difficult, field for further research.

Moulton's career and work offer the opportunity for an in-depth exploration of the way in which legal practice, technological innovation, scientific expertise and industrial development interacted in the law courts. They also throw valuable light on the close interplay between different social, scientific and professional networks: academic fraternities, political and legal élites, the circles of Fellows of the Royal Society, and the professional engineering associations (both the officially organized institutions and the informal but powerful community of London-based consulting engineers, with their offices clustered in the Victoria Street area). However, the relevant spaces of interplay could be extended even further, for example to the sheltered rooms of the Athenaeum. The relations between law, science and industry have already begun to receive more attention from historians of science and technology. Yet the profiles that tend to be chosen as probes into this multi-layered mesh of networks are those of scientists. What this note suggests is that other channels are well worth being pursued. Although Moulton stands out as a fascinating character, he was by no means the only one who deserves examination. Narrowing the list to the Fellows of the Royal Society, three equally valid candidates can be found among Moulton's colleagues and friends: Robert Romer

(FRS 1899), James Stirling (FRS 1902) and William Robert Bousfield (FRS 1916). These men, too, stood at the interface between science and the law and played an important role in some of the key events that helped to fashion the history of modern British industry.

NOTES

- 1 For the presentation speech and Marconi's lecture see *Nobel lectures. Including presentation speeches and laureates' biographies. Physics 1901–1921* (Elsevier, Amsterdam, 1967), pp. 191–221; and the Nobel website, http://nobelprize.org/nobel_prizes/physics/laureates/1909/index.html.
- 2 O. Lodge to J. J. Thomson, 15 January 1925, J. J. Thomson Papers, Cambridge University Library, MS ADD 7654, Correspondence, L4.
- 3 Campbell Swinton to W. H. Preece, 30 March 1896, IET Archive, London, William Preece's Archive, UK0108 NAEST 013.
- 4 A.R., 'Alan Archibald Campbell Swinton 1863–1930', *Proc. R. Soc. A* **130**, xiii–xv (1931); S. E. A. Landale (rev. Anita McConnell), 'Swinton, Alan Archibald Campbell', *Oxford dictionary of national biography* (Oxford University Press, 2004); A. A. Campbell Swinton, *Autobiographical and other writings* (Longmans, Green and Co., London, 1930); T. H. Bridgewater, *A. A. Campbell Swinton* (Royal Television Society, London, 1982), with a detailed list of his many publications.
- 5 No obituary was published in *Proceedings of the Royal Society*; C. Baker, *Sir William Preece, F.R.S.: Victorian engineer extraordinary* (Hutchinson & Co., London, 1976).
- 6 F.E.S., 'Henry Bradwardine Jackson 1855–1929', *Proc. R. Soc. A* **127**, vi–ix (1930); Rowland F. Pocock and G. R. M. Garratt, *The origins of maritime radio. The story of the introduction of wireless telegraphy in the Royal Navy between 1896 and 1900* (Science Museum, HMSO, London, 1972).
- 7 W. E. Eccles, 'John Ambrose Fleming 1849–1945', *Obit. Not. Fell. R. Soc.* **5**, 231–242 (1945–48); John A. Fleming, *Memories of a scientific life* (Marshall, Morgan & Scott, London, 1934). His collaboration with Marconi and his contribution to wireless telegraphy are examined in detail in Sungook Hong, 'Styles and credits in early radio engineering: Fleming and Marconi on the first transatlantic wireless telegraphy', *Ann. Sci.* **53**, 431–465 (1996).
- 8 J. A. Ratcliffe, 'William Henry Eccles 1875–1966', *Biogr. Mem. Fell. R. Soc.* **17**, 195–214 (1971). Another former employee of the Marconi Wireless Telegraph Co., Thomas Lydwell Eckersley, became a Fellow of the Royal Society; however, having joined the company in 1918, he did not collaborate directly with Marconi. J. A. Ratcliffe, 'Thomas Lydwell Eckersley 1886–1959', *Biogr. Mem. Fell. R. Soc.* **5**, 69–74 (1959).
- 9 No obituary was published in *Proceedings of the Royal Society*. For biographical information see Hugh Fletcher Moulton, *The life of Lord Moulton* (Nisbeth & Co., London, 1922); see also Theobald Mathew and Hugh Mooney, 'Moulton, John Fletcher', *Oxford dictionary of national biography*, www.oxforddnb.com/view/article/35132.
- 10 Wayne D. Cocroft, 'First World War explosives manufacture: the British experience', in *Frontline and factory: comparative perspectives on the chemical industry at war, 1914–1924* (ed. Jeffrey Allan Johnson and Roy MacLeod), pp. 31–46 (Springer, Dordrecht, 2006). See also Roy MacLeod, 'The chemists go to war: the mobilization of civilian chemists and the British war effort, 1914–1918', *Ann. Sci.* **50**, 455–481 (1993).
- 11 Obituary, *The Times*, 10 March 1921.
- 12 Maxim-Nordenfelt Guns and Ammunition Company, Ltd., and Hiram Stevens Maxim v. Anderson. See P. Fleury Mottelay, *The life and work of Sir Hiram Maxim* (John Lane, London, 1922).
- 13 Evelyn Hopkinson, *The story of a mid-Victorian girl* (privately published, Cambridge, 1928), pp. 45–46.

- 14 Among them, William Spottiswoode and J. F. Moulton, 'On the sensitive state of electric discharges through rarefied gases', *Phil. Trans. R. Soc.* **170**, 165–229 (1879); William Spottiswoode and J. Fletcher Moulton, 'On stratified discharges. VI. Shadows of striae', *Proc. R. Soc.* **32**, 385–387 (1881); 'On stratified discharges. VII. Multiple radiations from the negative terminal', *Proc. R. Soc.* **32**, 388–390 (1881).
- 15 As an expert in the field and a member of the Institution of Electrical Engineers, in the 1880s he participated in the works of the committees on the standardization of electrical units, strongly supporting the adoption of an international system. Moulton, *op. cit.* (note 9), p. 144; his contribution is also mentioned in Robert J. Strutt, *John William Strutt, Third Baron Rayleigh* (Edward Arnold, London, 1924), p. 123.
- 16 Moulton, *op. cit.* (note 9), pp. 99–100.
- 17 See, for example, R. E. Crompton, *Reminiscences* (Constable, London, 1928), p. 153; A. W. Ewing, *The man of room 40. The life of Sir Alfred Ewing* (Hutchinson & Co. London, 1939), pp. 128–129; Viscount Alverstone (Sir Richard Webster), *Recollections of Bar and Bench* (Edward Arnold, London, 1914), p. 195.
- 18 For a detailed account of the events see Anna Guagnini, 'Patent agents, legal advisers and Guglielmo Marconi's breakthrough in wireless telegraphy', *Hist. Technol.* **24**, 171–202 (2002).
- 19 Thompson to Lodge, 30 June 1897, University College Archive, Lodge Collection, MS ADD 89/104 I.
- 20 'Notes' (17 September), *Electrician* **39**, 665 (1897).
- 21 'Report to the Chief Engineer of the Government Telegraphs by Dr. Oliver Lodge. 2nd Part Completion of Dr. Oliver Lodge's Report', June 1900, p. 4, Public Record Office, ADM 116/570, 30597.
- 22 'Report received from Mr J. Fletcher Moulton, FRS, Q.C. of King's Bench Walk, E.C., 6 March 1897', Document enclosed in 'The Wireless Telegraph and Signal Company, Ltd. Prospectus', Oxford, Bodleian Library, MS. Marconi, His 44. (Here and below I use the document references of the Catalogue of the Marconi Archive, now superseded by the new catalogue prepared by the Bodleian Library, Oxford.)
- 23 Marconi, draft letter to H. R. Kempe, Oxford, Bodleian Library, MS. Marconi, His 208.
- 24 Major Samuel Flood Page to Marconi, 2 January 1901, Oxford, Bodleian Library, MS. Marconi, His 210; Fleming to Marconi, 20 February 1901, Oxford, Bodleian Library, MS. Marconi, His 248–249.
- 25 John Fletcher Moulton, untitled note concerning Messrs Betts' opinion, Oxford, Bodleian Library, MS. Marconi, His 181. Moulton was asked to examine the document prepared by Marconi's American solicitors, Betts, Betts, Sheffield & Betts, of New York; see Moulton, 'Report for Betts and also Comparison between the 1st and 3rd edition of the works of Hertz', Oxford, Bodleian Library, MS. Marconi, His 208.
- 26 Fleming to Flood Page, 23 November 1900; Fleming to Marconi, 17 February 1901; Flood Page to Marconi, 19 February 1901, Oxford, Bodleian Library, MS. Marconi, His 208.
- 27 In 1911 Lodge's wireless telegraphy patents were purchased by the Marconi Wireless Telegraph Co. for £20 000 plus a (nominal) position on the board of the company's directors.
- 28 As his son pointed out, 'One special feature of his character was his great reticence and regard for any confidence imposed in him, and even in his own family he never referred to anything that had been said to him as to which there could be any desire of secrecy. Further, he kept no diary and hardly any letters, considering the fire the only trustworthy guardian of confidences'. See Moulton, *op. cit.* (note 9), p. 147.